Centrifugal Pump P-102 — Technical Manual

Model: CP-500X | Manufacturer: HydroFlow Industries

1. General Description

Pump P-102 is a horizontal centrifugal pump used to circulate cooling water in the secondary loop of the plant. Rated flow is 250 m3/h at a differential pressure of 4.2 bar. The pump is driven by a 75 kW three-phase induction motor running at 2950 RPM, coupled via a flexible elastomeric coupling.

2. Operating Parameters

Parameter	Normal Range	Alarm Threshold
Bearing temperature (drive end)	40-65 C	> 80 C
Bearing temperature (non-drive end)	40-65 C	> 80 C
Vibration (RMS velocity)	0-4.5 mm/s	> 7.1 mm/s
Discharge pressure	3.8-4.5 bar	< 3.2 bar or > 5.0 bar
Motor winding temperature	< 120 C	> 155 C
Seal chamber pressure	1.5-2.0 bar	< 1.0 bar

3. Vibration and Overheating — Common Causes

Abnormal vibration on P-102 is most frequently caused by one of the following: (a) impeller imbalance due to erosion or debris buildup, typically producing vibration at 1x running speed; (b) coupling misalignment between pump and motor shafts, producing 2x running speed vibration and often accompanied by elevated bearing temperature; (c) bearing wear or lubrication starvation, producing high-frequency vibration and a steady rise in bearing temperature over several days; (d) cavitation at the suction side, caused by insufficient NPSH, which produces a characteristic crackling noise and erratic, broadband vibration.

Overheating of the bearings is most commonly linked to lubrication issues (low oil level, degraded grease, or wrong grease type) or to misalignment that increases radial load on the bearings. Motor winding overheating is usually caused by voltage imbalance across phases, prolonged operation below minimum flow, or a partially blocked cooling fan.

4. Recommended Inspection Sequence

1. Check current sensor readings for bearing temperature, vibration and discharge pressure against the thresholds in Section 2.
2. Review maintenance history for recent lubrication or alignment work.
3. If vibration is at 1x running speed, inspect impeller for wear/debris.
4. If vibration is at 2x running speed with high bearing temperature, check shaft alignment with a laser alignment tool (tolerance: 0.05 mm).
5. If bearing temperature rises steadily without a vibration signature change, inspect lubrication (oil level, grease condition).
6. If discharge pressure is low and vibration is erratic, check suction strainer for clogging and verify NPSH available vs. NPSH required.

5. Maintenance Intervals

Lubrication: re-grease bearings every 2000 operating hours. Alignment check: every 6 months or after any coupling work. Vibration baseline survey: every 3 months. Full bearing replacement: every 25,000 operating hours or on condition.